(8) **Assignment;**

Problem-1:

(1)

ref. ramp = 1.953 V/ms

total possible count = $2^8 = 256$

$\therefore$ ~~total~~ time $= \frac{1}{50} \times 2^8$

$= 5.12$ ms

~~Ramp sto~~ Max voltage $= 5.12 \times 1.953$

$= 9.99936$ V

Quantization step size $= \frac{V_{max} - V_{min}}{L}$

$= \frac{\cancel{9.96}\ 9.99936}{2^8}$

$= \cancel{0.0389V} = 0.03906$ V (Ans)

~~= 39mV (Ans)~~

(ii)

$V_{max} = 9.99936$

No. of counts for $V_{max} = 2^8$

" " " " $3.91 = \frac{2^8}{9.99936} \times 3.91$

$= 100.10$

In 8-bit binary: 01100100 (Ans)

(iii)

Maximum conversion time, $t = 2^8 \times \frac{1}{50}$

$= \cancel{5.12ms}$ 5.12 ms (Ans)

Problem-2:

(i)

Given,

Frequency = 100 kHz

Time = 40.96 ms

In 1sec of clock pulse = ~~10×1~~ 100×10^3 count

" 40.96ms " " $= 100 \times 10^3 \times 40.96 \times 10^{-3}$

$= 4096$ count (Ans)

(ii)

$4096 = 2^N$

$\therefore N = \log_2(4096) = 12$ (Ans)

P.T.O

(iii)

No. of counts for reaching 5V = 4096

" " " " " 2.41V = $\frac{4096}{5} \times 2.41$

= 1974.272

Required time = $\frac{1}{100} \times 1974.272$

= 0.019741

= 19.74 ms (Ans)

(iv)

No. of count = 1974.

In binary = 0111 1011 0110

~~Ans to the Ques No 03~~

Problem-3:

(i)

for single slope,

$V_{ref} = 5V$

maximum convertion time = $\frac{2^{R}}{10^{6}}$ = 4.096 ms

convertion time for 4.5V = $\frac{4.096}{5} \times 4.5$ = 3.685 ms

for dual slope,

Time required = (4.096 + ~~3.8~~ 3.685) ms

= ~~7.8~~ 7.781 ms (Ans)

(i)

The no. of count for both cases one is same.

$$No.\ of\ count = \cancel{3.68}\ 3.685 \times 10^{-3} \times frequency$$

$$= 3.6855 \times 10^{-3} \times 1 \times 10^{6}$$

$$\approx \cancel{3685}\ 3686$$

Digital Output Code = 1110 0110 0110 (Ans)

(iii)

Dual Slope ADC is generally more suitable for high accuracy on noisy environments, because,

i) it integrates the input for a fixed time, so random and high frequency noise is averaged out.

ii) The result doesn't depend on RC constant of capacitor which is suspectible to change due to temperature, so the result is stable

iii) If the result is set to match the ~~single~~ main cycle, it cancels power line interfer

In contrast, single slope ADC is faster but the dependence on ~~A~~ RC ~~cons~~ value and noise makes it prone to error.

P.T.O